SAMPLE

Earth's Dynamic Systems: Earth's Magnetic Field

The Geodynamo

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Introduction

There is a force around you right now that you cannot see, hear, or feel, one that has guided sailors across oceans, drawn curtains of light over the poles, turned aside a lethal wind from the Sun for four billion years, and left its diary written in the rocks of the seafloor. It is Earth's magnetic field: an invisible structure of field lines running from poles near, but not at, the geographic ends of the planet, looping tens of thousands of kilometers out into space and returning, generated by a churning ocean of molten iron nearly three thousand kilometers beneath your feet. A compass needle settles along it. Birds read it. Every piece of iron-cooled lava and every fired clay pot on the planet carries a faint memory of it, and the planet's habitability, the preservation of its atmosphere from the Sun's steady erosion, is plausibly one of its quiet works.

This book is an account of that field for the general reader: what it is and how it is shaped, the engine in the core that generates it, the centuries of human use and study from the compass to the satellite, the restless changes it undergoes year to year and eon to eon, the shield it throws around the planet and the lights and storms that shield makes possible, the way its history is recorded in rock and read by scientists, and the puzzles it still poses, including the great one, why the field occasionally flips its poles entirely, a question the companion volume on reversals takes up in full. No mathematics is needed; the story is told in plain narrative, and it is a story of two worlds at once, the glowing iron engine inside the planet and the vast invisible architecture it builds in space.

The field deserves a book because it is the planet's most personal force, the one that belongs to the Earth's own life rather than to the general laws of physics. Gravity, sunlight, and chemistry are the same everywhere; a magnetic field like ours belongs to a world with a molten metal heart, convecting and spinning, and the Earth has one because it is large enough to still be hot inside and young enough to have kept its spin. The field is thus a window into the deep interior, the only one that reaches us in real time, and the record of its past, read from rocks, is one of the great archives of planetary history. The chapters that follow open that window and read that archive, and they end where the subject ends, with a planet whose invisible shield is weakening in one hemisphere, drifting in its poles, and behaving, in the words of every honest summary, in a way that keeps the scientists curious.

The itinerary, for readers who like to know the country before they walk it. Two historical chapters open the account, the compass and the invisible architecture; two interior chapters follow the engine, the core's dynamo and the restless changes it shows at the surface; then the field's works, the shield in space, the aurora, the rock record, the storms; and the closing chapters gather the biological compasses, the modern measurement enterprise, and the field's role in keeping the planet habitable. The reversals appear here as one thread in the whole cloth, treated at the length a general account allows, and the companion volume on geomagnetic reversals picks up that thread and follows it into the details of the flip itself, should the reader want the full story of the compass turning around.

Chapter 1: A Force Older Than Its Discovery

The story of the field begins with a stone and a needle. In the ancient world, in more than one place, certain dark stones were noticed to attract iron, and by themselves to swing, when floated or hung, toward the same direction, north and south; the Chinese made instruments of the property early, the lodestone spoon on a bronze plate, and the mariner's needle, when it reached the Mediterranean, changed navigation forever. A ship that could hold its course under clouded skies crossed oceans it had no business crossing, and the age of exploration, with all its consequences, sailed on a magnetized sliver of iron that pointed, for reasons no sailor knew, at an invisible something none of them could name.

The first honest theory of the field came from a court physician in the year 1600, in a book whose title, in translation, is On the Magnet, and its author's argument was radical for its day: the Earth itself, William Gilbert wrote, is a great magnet. He tested the idea with a sphere of lodestone, a little model Earth, and showed that a small compass needle, moved about the sphere, tilted and pointed the way compasses behave on the globe, dipping as they do at high latitudes; the field's strange behaviors, the tilt, the dip, the variation from true north, all belonged to the planet's own magnetism, not to some attractive star at the north or a mountain of iron in the arctic, as earlier theories had held. Gilbert's conclusion, that the globe is a magnet throughout, was wrong in mechanism but right in essence, and it founded the science this book tells.

The problem Gilbert left was the mechanism, and it took two more centuries to state it correctly. The Earth is not a solid lodestone: its interior is far too hot to hold permanent magnetism, and the field's source had to be something dynamic, an engine, not a rock. The clue came from electromagnetism in the nineteenth century, when it was

shown that moving electric currents create magnetic fields, and the modern question was framed: the planet's field must be generated by electric currents, flowing somewhere inside it, and the currents must be maintained by motion. The candidate location was the core, once the planet's deep interior was mapped by seismology in the early twentieth century, and the candidate motion was the churning of liquid iron, an ocean of metal thousands of kilometers deep, convecting under the heat of the planet's interior and stirred by the planet's rotation, an engine whose full theory, as later chapters tell, was one of the great achievements of the physics of the last century.

A note on the compass's actual behavior belongs here, because the history's most practical fact, the needle's disagreement with true north, was the first evidence that the field is not simple. Navigators early on noticed that the needle points not at the geographic pole but at a wandering magnetic one, and that the disagreement, called the declination, differs from place to place and drifts with the years, so much so that old charts carried its measured values for their waters, and a science of magnetic navigation, with its expeditions to find the magnetic poles themselves, grew up alongside the pure science. The field, the navigators knew before the physicists did, is restless, and its restlessness, tracked now by satellites and observatories and chronicled in every later chapter, is one of this book's recurring characters.

The instrument's early forms repay a closer look, because they show how much was achieved with almost nothing. The Chinese south-pointing spoon, a lodestone carved to sit on a polished bronze plate, was a divination instrument as much as a tool of direction, and its successor, the suspended needle, reached European sailors by the twelfth century and appears in the sailors' manuals of the age as a wonder used without shame; the needle was magnetized by stroking it with lodestone, floated on straw or hung on a silk thread, and its cardinal virtue, that it worked in fog, cloud, and night, transformed the Mediterranean from a coastal highway into an open sea. Within two centuries the Atlantic had been crossed, the Indian Ocean's monsoon routes were being flown by needle, and the planet's unification under sail, with all its consequences for the world's peoples, rested on a sliver of stroked iron, an apparatus no more complicated than a needle and a cork, guided by a force the users could not begin to explain.